Yashi Ajay Dubey

Junior Solution Architect · AI Systems · Research

yashidubey0102@gmail.com · +91 8208429420 · Pune, Maharashtra, India

GitHub: github.com/bellatrix01 · LinkedIn: linkedin.com/in/yashidubey · Portfolio: bellatrix01.github.io

RESEARCH INTERESTS

AI Systems & LLM Infrastructure · Agentic AI & Multi-Agent Systems · Distributed Systems & Cloud Architecture · Blockchain & Decentralized Systems · AI Reliability, Observability & Governance

EDUCATION

Bachelor of Technology in Computer Science & Engineering Sep 2020 – Jun 2024

Dr. Babasaheb Ambedkar Technological University (DBATU) · Pune, Maharashtra

GPA: 3.49 / 4.0 | Batch Topper – First Year (9.89 CGPA, highest in cohort) | Dean's List – All Academic Years

Relevant Coursework: Data Structures & Algorithms, Distributed Systems, Machine Learning, Cloud Computing, Database Management Systems, Operating Systems, Computer Networks, Artificial Intelligence

Senior Secondary Education (Class XII) 2020

Maharashtra State Board of Secondary and Higher Secondary Education

Percentage: 83% | Stream: Science (Mathematics, Physics, Chemistry)

Secondary School Education (Class X) 2018

Priyadarshani High School, Maharashtra

Percentage: 85.60%

RESEARCH PROJECTS

LLM Inference Optimization Independent Research

Python · PyTorch · Hugging Face Transformers · Profiling Tools *Focus: Efficient AI Systems · LLM Optimization*

A research-oriented prototype investigating latency and memory efficiency of transformer-based LLMs during inference. Explores system-level optimizations such as KV cache reuse and model quantization under varying workload conditions, particularly in long-context scenarios.

- Implemented KV cache mechanisms to eliminate redundant attention computations during sequential token generation
- Profiled system behavior to identify memory-bound bottlenecks in transformer inference pipelines
- Analyzed trade-offs between model efficiency and output quality under different optimization strategies

Research Question: How can system-level optimizations improve scalability and real-time performance of large language models in resource-constrained, latency-sensitive environments?

Compass AI — Agentic Monitoring & Governance for AI Systems Associated with Johnson Controls

Python · LLMs · Vector Databases · Cloud Services · Distributed Logging & Monitoring *Focus: Agentic AI · ML Systems · AI Safety*

A research-oriented prototype addressing the lack of observability, reliability, and governance in large-scale LLM-integrated production systems. Introduces an agentic supervisory layer that continuously monitors AI behavior, evaluates output quality, and autonomously triggers corrective actions (rerouting, retries, fallback strategies).

- Designed autonomous observer, evaluator, and decision-making agents for closed-loop AI control
- Implemented closed-loop control mechanisms for LLM system reliability and fault tolerance
- Explored metrics for hallucination detection, output drift, and policy violation monitoring
- Integrated memory and longitudinal logging layers for behavioral analysis over time

Research Question: How can agentic control loops improve robustness, trust, and fault tolerance in LLM-driven production systems?

SentinelAI — AI-Driven Cybersecurity Threat Detection Independent

Python · Anomaly Detection · Event-Driven Cloud Architecture *Focus: AI Security · Real-Time Systems*

An AI-assisted threat detection framework leveraging anomaly detection and event-driven cloud architecture. Implemented a Ripple Network Model to propagate threat intelligence signals across distributed services in near real-time.

- Designed scalable architecture for low-latency event processing and intelligent alert dissemination
- Implemented Ripple Network Model for distributed threat intelligence propagation

Decentralized Audit & Provenance System for Intelligent Services DBATU · Pre-Final Year

Blockchain (Permissioned Ledger) · Smart Contracts · Cryptographic Hashing · Off-chain Storage *Focus: Blockchain · Distributed Ledgers*

Investigates blockchain-based architectures to provide auditability, provenance, and tamper-resistance for decisions generated by distributed software systems — ensuring transparency and accountability in environments where decisions must be verifiable over time.

- Designed a permissioned blockchain architecture for enterprise environments
- Implemented immutable logging of system events and decision metadata
- Used smart contracts to enforce validation and access-control policies
- Enabled off-chain storage with on-chain integrity verification

Research Question: How can decentralized ledgers improve trust, accountability, and fault diagnosis in large-scale distributed systems without significant performance overhead?

Multi-Agent LLM System for Decision Automation Independent

Python · LLM APIs · Prompt Chaining · Observability Tools

Built a multi-step agent system using LLM APIs for task planning and execution. Implemented prompt chaining, tool usage, and response validation. Designed evaluation pipeline with logging and observability for production readiness.

Production-Grade AWS VPC Architecture Independent

AWS VPC · EC2 · Load Balancers · NAT Gateway · Auto Scaling

Designed and implemented a multi-Availability Zone AWS VPC with public/private subnets, load balancers, NAT gateways, and Auto Scaling groups for fault tolerance and variable traffic demands.

PATENTS & PUBLICATIONS

WEB3NEXUS — Next-Generation Web Application Network Published · July 12, 2024

India Patent · Application No: 202421039913 · Publication No: 28/2024

Presents a decentralized web application framework designed to improve scalability, interoperability, and trust in Web3-based systems. The proposed architecture emphasizes secure data exchange, decentralized control, and system-level reliability for next-generation web platforms.

Research Paper

"Exploring Blockchain Solutions for Enhancing Trust in IoT Environments"

PROFESSIONAL EXPERIENCE

Johnson Controls — Pune, India

Junior Solution Architect (Rotational Role)

Dec 2025 – Present

- Selected for rotational Junior Solution Architect role after ~1 year of full-time industry experience, reflecting early trust in architectural responsibilities
- Designed and deployed AI-assisted voice and agent workflows integrating enterprise platforms with intelligent automation (Copilot-style assistance)
- Architected event-driven and serverless workflows integrating NICE CXone, Cognigy, and Five9 (SaaS)
- Implemented real-time decision systems for call routing, intent handling, and agent assistance — functionally agentic pipelines
- Participated in architectural discussions on distributed systems, API orchestration, and platform interoperability

IT Analyst

Jan 2025 – Nov 2025

- Engineered workflow automation and logic pipelines simulating intelligent decision trees, optimizing call routing and ticket lifecycle management
- Built and integrated RESTful APIs to connect NICE CXone, Salesforce (SFDC), and internal backend services
- Assisted in monitoring, troubleshooting, and optimizing production workflows in live enterprise environments
- Mentored two interns on enterprise platform onboarding and technical training

Graduate Software Engineer

Jan 2024 – Dec 2024

- Administered and maintained enterprise-grade NICE CXone systems ensuring optimal performance and reliability
- Developed and enhanced Studio scripts to automate workflows and improve system functionality
- Collaborated with cross-functional teams on system updates and deployments

TECHNICAL SKILLS

Programming: Python, Java, C++, JavaScript
AI / LLM: Prompt Engineering, LLM APIs, LangChain, RAG (foundational), Agentic Systems, KV Cache Optimization, Quantization
Backend & Systems: RESTful APIs, Spring Boot, Node.js, Distributed Systems, Event-Driven Architecture
Cloud & Infrastructure: AWS (VPC, EC2, Load Balancers, Auto Scaling, NAT), Azure OpenAI, Data Lake
Blockchain: Permissioned Ledgers, Smart Contracts, Cryptographic Hashing, Web3 Architecture
Tools & Platforms: Git, GitHub, Postman, NICE CXone, Cognigy, Five9, VS Code, Linux (basic)
Core CS: Data Structures & Algorithms, OOP, Database Systems, Operating Systems, Computer Networks

AWARDS & ACHIEVEMENTS

- Dean's List (All Academic Years) — Recognized for consistent academic excellence throughout the undergraduate program
- Batch Topper, First Year B.Tech — Secured 9.89 CGPA, highest in cohort
- Smart India Hackathon 2022 – National Finalist — Developed a technical solution for a Government of India problem statement at the national level
- Best Innovation Award (3rd Year) — Awarded for Blockchain + AI-based system design at the university project exposition
- Johnson Controls Innovation Challenge – Finalist — Led a team of 8 to design and prototype an enterprise AI solution during internship
- 2nd Place – Inter-College Website Development Competition

LEADERSHIP & RESPONSIBILITIES

Project Lead — Undergraduate Thesis & Capstone

Led a team of four during pre-final and final year project development. Designed technical architecture, implemented core system components, and presented end-to-end workflow during evaluations and project demonstrations.

Technical Lead — BitsToBytes (CSE Technical Club)

Coordinated departmental hackathons, coding competitions, and technical events. Managed planning, scheduling, and execution of large-scale student engagement initiatives within the Computer Science department.

Intern Mentor — Johnson Controls

Guided and trained multiple batches of interns on enterprise platforms including NICE CXone. Mentored two direct interns in the first year; currently supervising four interns supporting onboarding, technical training, and project execution.

Technical Session Speaker — Blockchain & Web3 Systems

Delivered technical sessions to AI-focused student batches, presenting blockchain architectures and decentralized application workflows.

Academic Mentor — Secondary School Students (Grades 6–9)

Taught Mathematics and English during undergraduate studies to support self-financing and develop instructional and mentoring skills.

ADDITIONAL

- District-level Karate Player during school years. Elementary and Intermediate Drawing Examination holder (B+ Grade)